Infinite limits :

Definition 3 : let f be a function defined on both sides of a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a, but not equal to a.

" ∞ " is not a number

$\lim_{x \to a} f(x) = \infty$ can be read " $f(x)$ increases without bound as x approaches a "

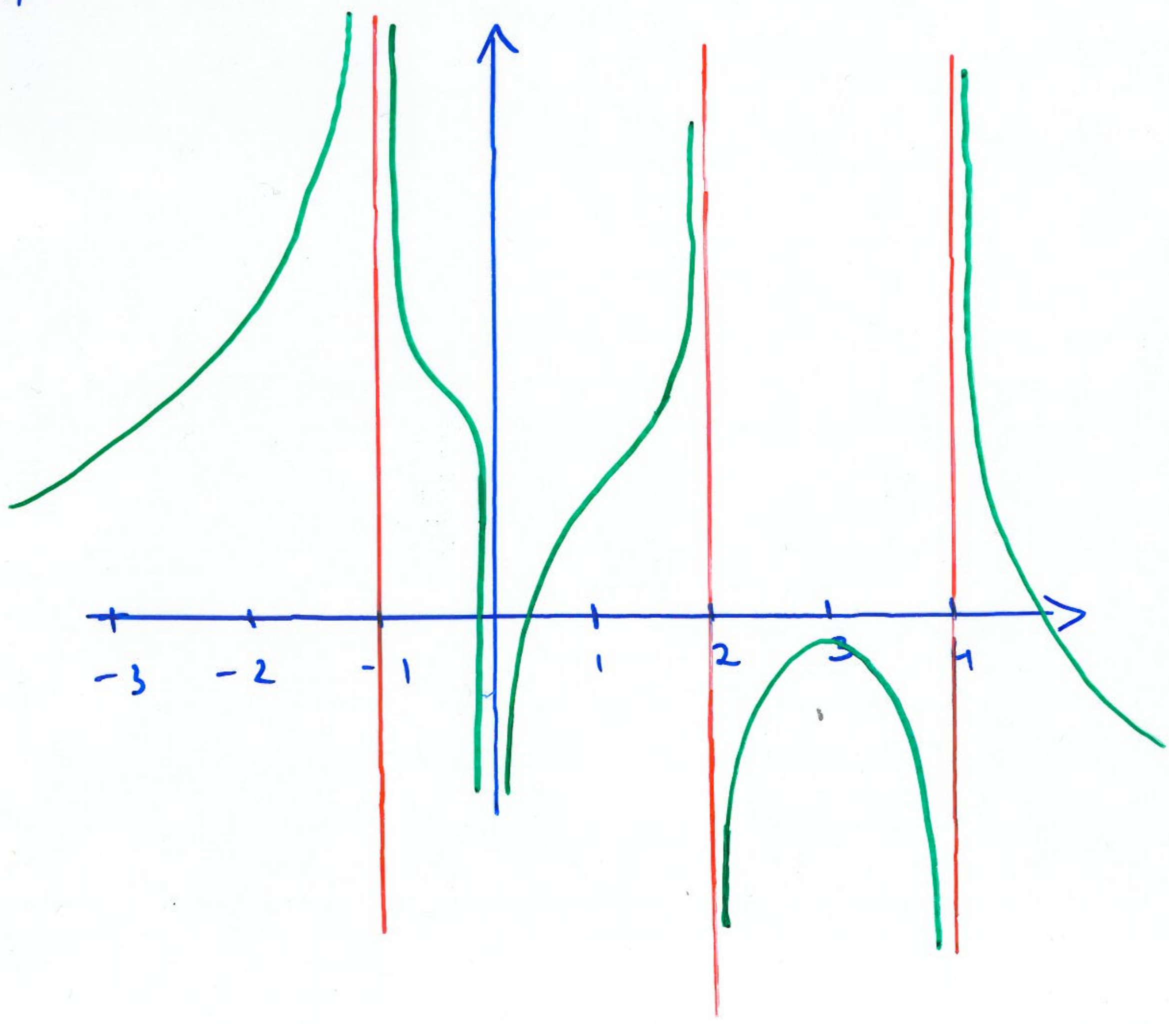
Definition 4: Let f be defined on both sides of a, except possibly at a itself.

Then

$$\lim_{x \to a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a, but not equal to a.

" $f(x)$ decreases without bound as x approaches a "



$\lim_{x \to -1} f(x) = +\infty$ since $\lim_{x \to -1^+} f(x) = \lim_{x \to -1^-} f(x) = +\infty$

$\lim_{x \to 0} f(x) = -\infty$ since $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = -\infty$

$\lim_{x \to 2^-} f(x) = +\infty$ and $\lim_{x \to 2^+} f(x) = -\infty$

therefore $\lim_{x \to 2} f(x)$ does not exist

similarly, $\lim_{x \to 4} f(x)$ does not exist

$\left(\lim_{x \to 4^-} f(x) \neq \lim_{x \to 4^+} f(x) \right)$

In writing

NB: $\lim_{x \to a} f(x) = \pm\infty \neq \lim_{x \to a} f(x)$ does not exist

Example 6: Find

(a) $\lim_{x \to 2} \frac{x+2}{x^2-4}$

(b) $\lim_{x \to 1} \frac{1}{|x-1|}$

(c) $\lim_{x \to 0} \frac{1}{x}$

Example 6 :

(a) $$\lim_{x \to 2} \frac{x+2}{x^2-4} = \lim_{x \to 2} \frac{x+2}{(x-2)(x+2)} = \lim_{x \to 2} \frac{1}{x-2}$$

$$\lim_{x \to 2^-} \frac{1}{x-2} = -\infty \quad ; \quad \lim_{x \to 2^+} \frac{1}{x-2} = +\infty$$

$\lim_{x \to 2} \frac{x+2}{x^2-4}$ does not exist

(b) $$\lim_{x \to 1} \frac{1}{|x-1|} = +\infty$$

since $\lim_{x \to 1^-} \frac{1}{|x-1|} = \lim_{x \to 1^-} \frac{1}{1-x} = +\infty$

and $\lim_{x \to 1^+} \frac{1}{|x-1|} = \lim_{x \to 1^+} \frac{1}{x-1} = +\infty$

(c) $$\lim_{x \to 0^+} \frac{1}{x} = +\infty \quad ; \quad \lim_{x \to 0^-} \frac{1}{x} = -\infty$$

$\lim_{x \to 0} \frac{1}{x}$ does not exist

Definition 5: The line $x=a$ is called a vertical asymptote of the curve $y=f(x)$ if at least one of the following statements is true:

$$\lim_{x\to a} f(x)=\infty \qquad \lim_{x\to a^-} f(x)=\infty \qquad \lim_{x\to a^+} f(x)=\infty$$

$$\lim_{x\to a} f(x)=-\infty \qquad \lim_{x\to a^-} f(x)=-\infty \qquad \lim_{x\to a^+} f(x)=-\infty$$

Remark: If $a \notin D_f$ then $x=a$ could be a vertical asymptote.

Example 7: Find the vertical asymptote(s) of f, if any.

(a) $f(x)=\tan x$; (d) $f(x)=\dfrac{1}{x}$

(b) $f(x)=\ln x$; (e) $f(x)=\dfrac{x+2}{x^2-4}$

(c) $f(x)=\csc x$ f) $f(x)=\dfrac{e^x-1}{x}$

Example 7 :

(a) $D_f = \mathbb{R} \setminus \left\{ \frac{\pi}{2} + 2k\pi \, , \, k \in \mathbb{Z} \right\}$

on $[0, 2\pi]$, $\cancel{D_f} [0 , \frac{\pi}{2}) \cup (\frac{\pi}{2} , \frac{3\pi}{2}) \cup (\frac{3\pi}{2} , 2\pi]$

let $a = \frac{\pi}{2}$:

$$\lim_{x \to \frac{\pi}{2}^-} \tan x = \lim_{x \to \frac{\pi}{2}^-} \frac{\sin x}{\cos x} = +\infty$$

$$\lim_{x \to \frac{\pi}{2}^+} \tan x = \lim_{x \to \frac{\pi}{2}^+} \frac{\sin x}{\cos x} = \lim_{x \to \frac{\pi}{2}^+} \frac{1}{\cos x} = -\infty$$

$x = \frac{\pi}{2}$ is a V. A

let $a = \frac{3\pi}{2}$:

$$\lim_{x \to \frac{3\pi}{2}^-} \tan x = \lim_{x \to \frac{3\pi}{2}^-} -\frac{1}{\cos x} = +\infty$$

$$\lim_{x \to \frac{3\pi}{2}^+} \tan x = \lim_{x \to \frac{3\pi}{2}^+} -\frac{1}{\cos x} = -\infty$$

$x = \frac{3\pi}{2}$ is a V.A

Hence any $x = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$

is a V. A

(b) $D_f = (0, +\infty)$

<u>let $a = 0$</u>

$\lim_{x \to 0^+} \ln x = -\infty$, so $x = 0$ is the only V.A.

(c) $D_f = \mathbb{R} \setminus \{k\pi, k \in \mathbb{Z}\}$

on $[0, 2\pi]$, $D_f = (0, \pi) \cup (\pi, 2\pi)$

<u>$a = 0$</u>

$\lim_{x \to 0^+} \csc x = \lim_{x \to 0^+} \frac{1}{\sin x} = +\infty$

$x = 0$ is a V.A

<u>$a = \pi$</u>

$\lim_{x \to \pi^-} \frac{1}{\sin x} = +\infty$; $x = \pi$ is a V.A

$\lim_{x \to \pi^+} \frac{1}{\sin x} = -\infty$

<u>$a = 2\pi$</u>

$\lim_{x \to 2\pi^-} \frac{1}{\sin x} = -\infty$; $x = 2\pi$ is a V.A

any $x = k\pi$, $k \in \mathbb{Z}$ is a V.A

(d) $D_f = (-\infty, 0) \cup (0, +\infty)$

$a = 0$

$\lim\limits_{x \to 0^+} f(x) = +\infty$; $\lim\limits_{x \to 0^-} f(x) = -\infty$

$x = 0$ is a V.A (the only V.A)

(e) $D_f = \mathbb{R} \setminus \{2, -2\}$

from Eg 6 (a)

$x = 2$ is a V.A

$a = -2$:

$\lim\limits_{x \to -2} f(x) = \lim\limits_{x \to -2} \frac{1}{x-2} = \frac{1}{-2-2} = -\frac{1}{4}$

$x = -2$ is not a V.A

(f) $D_f = \mathbb{R} \setminus \{0\}$

$\lim\limits_{x \to 0^+} f(x) = 1$; $\lim\limits_{x \to 0^-} f(x) = +1$ (from 2.1)

So $x = 0$ is not a V.A

Remark : $a \notin D_f$ is not sufficient to call $x = a$ a verticale asymptote. You must always test the limit.